

Shot Size / Cushion Study in Plastic Injection Molding

A technical reference for processors, engineers, technicians, mold makers, quality engineers, and manufacturing leaders

Shot size / cushion terminology on a reciprocating-screw press

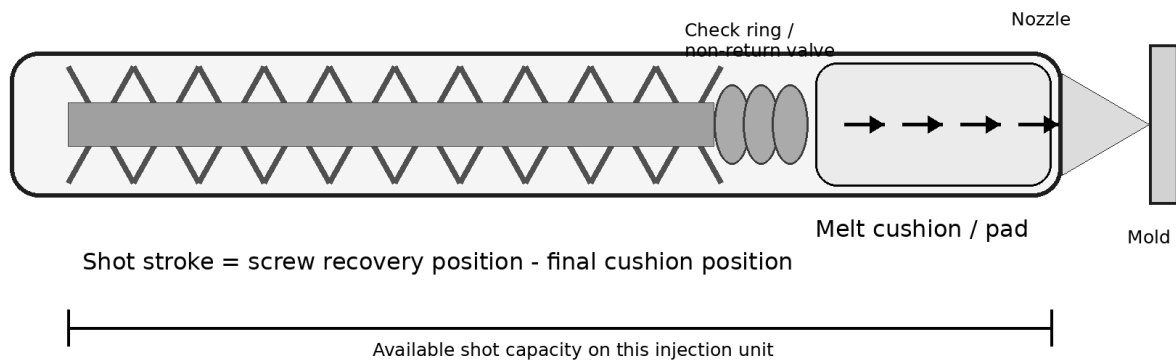


Figure 1. Core terminology: shot stroke, transfer, final cushion, and melt pad.

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Accuracy and review status: This document is written from authoritative public references, scientific-molding practice, and production-oriented reasoning. It has not been formally peer-reviewed by named SPE fellows, OEM application engineers, or a resin supplier for a specific plant, resin, machine, or mold. Before release as a controlled work instruction, have it reviewed by your internal process engineering lead, tooling lead, quality representative, machine OEM representative when applicable, and resin/hot-runner supplier for the specific asset.

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Executive Summary

A shot size / cushion study is a controlled injection-molding experiment used to establish the screw recovery position, transfer position, and residual melt cushion needed to fill the mold in a repeatable, pressure-transmitting, non-bottomed-out condition. Its immediate deliverables are not just a “shot size number.” The deliverables are a validated fill-only volume, a transfer point near the intended fill percentage, a final cushion control band, repeatable fill time, repeatable peak pressure, and evidence that pack/hold pressure can be transmitted to the cavity without running out of melt in front of the screw.

The study is one part of a larger scientific-molding sequence. The relative-viscosity curve is normally used to select a robust fill rate. A shot size / cushion study then uses that selected fill rate to determine the correct volume and transfer target. A gate seal study, pressure-loss study, cavity balance study, and process-window study usually follow or support it. Treating all of these as the same test leads to false conclusions.

Cold runner and hot runner molds must be handled differently. In a cold runner mold, each cycle consumes part volume plus sprue, runner, cold sub-runners, and gates. In a hot runner mold, the manifold melt is not ejected every shot, but its compressible volume, temperature balance, valve-gate timing, tip condition, shear history, and controller stability strongly influence the apparent cushion, pressure profile, cavity balance, and fill repeatability.

The correct cushion is mold-, material-, machine-, and application-specific. Generic rules such as 3-6 mm or 5-10% of shot stroke can be useful startup boundaries, but they are not substitutes for data. Too little cushion risks screw bottoming, loss of pack pressure transmission, short shots, dimensional drift, and machine damage. Too much cushion increases residence time, heat history, material degradation risk, and process variability.

Control variable	What the study should establish	Why it matters on the floor
Fill-only shot volume	Volume needed to fill the feed system and cavities to a controlled partial-fill target, commonly about 95-98% of molded volume before pack.	Separates fill from pack and prevents using hold pressure to hide an incorrect transfer point.
Transfer position / V-P switchover	Screw position where the machine changes from velocity-controlled filling to pressure-controlled pack/hold.	Prevents underfill, flash, overpacking, pressure spikes, and dimensional instability.
Final cushion / minimum cushion	Residual melt pad after injection and hold, tracked over production shots.	Confirms pressure transmission, check-ring performance, and shot repeatability.
Fill time and peak pressure	Repeatable response variables under identical settings.	Indicates viscosity shifts, venting issues, pressure limitations, and process drift.
Part weight and cavity balance	Weight and short-shot fill pattern at the selected fill-only state and final packed process.	Connects machine settings to molded-part quality, not just machine movement.

1. Definition and Overview

1.1 Precise technical definition

A shot size / cushion study is a structured, controlled molding study that determines the relationship between injected screw stroke, transfer position, residual melt cushion, fill-only part completion, pressure response, and molded part response for a specific mold, material, machine, screw/non-return valve condition, and thermal state. The study is normally executed after the machine, mold, resin, water circuits, hot-runner controller if present, and auxiliary equipment have reached a repeatable operating condition.

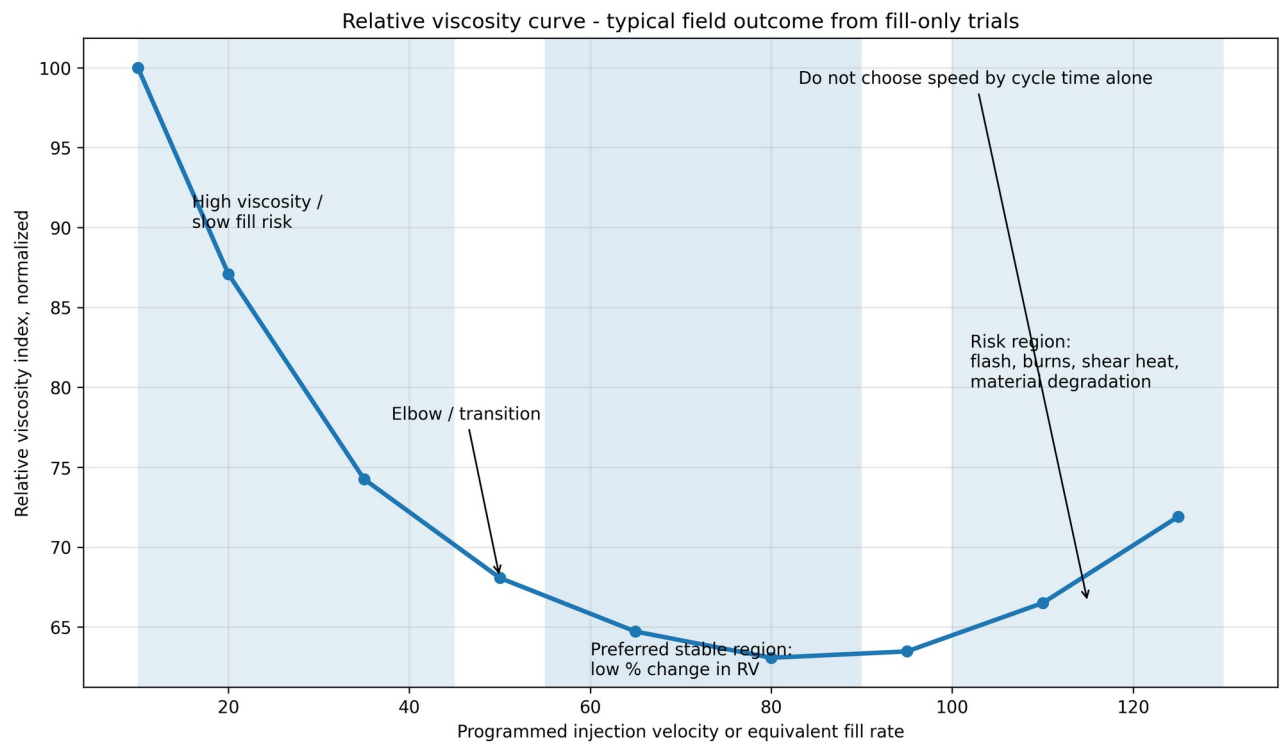
The study has two technical objectives. First, it establishes a fill-only volume and transfer position that leave the cavity almost full, but not packed, at the end of velocity-controlled injection. Second, it establishes a residual cushion that is large enough to maintain pressure transmission during pack/hold, yet small enough to avoid excessive residence time, degradation, and compressible melt inventory.

The study is machine-position based, but the decision criteria are not machine-position only. A correct result is validated with fill time, peak pressure, short-shot appearance, part weight, cavity balance, final cushion, dimensional checks, cosmetic evaluation, and, when available, nozzle or cavity pressure data.

Basic volumetric relationship. For a reciprocating screw, approximate injected volume from stroke is calculated as:

$$\text{Injected volume} = (\pi \times \text{screw diameter}^2 / 4) \times \text{screw travel}$$

Use this only as an engineering estimate. Real injected mass depends on melt density, compressibility, check-ring leakage, decompression, material slip, actual screw diameter, hydraulic response, and the machine control architecture.



Illustrative data. Actual values must be calculated from the press, nozzle pressure, or cavity-pressure data under controlled fill-only conditions.

Figure 2. Typical relative-viscosity-curve outcome used to choose a robust fill rate before finalizing shot size and cushion.

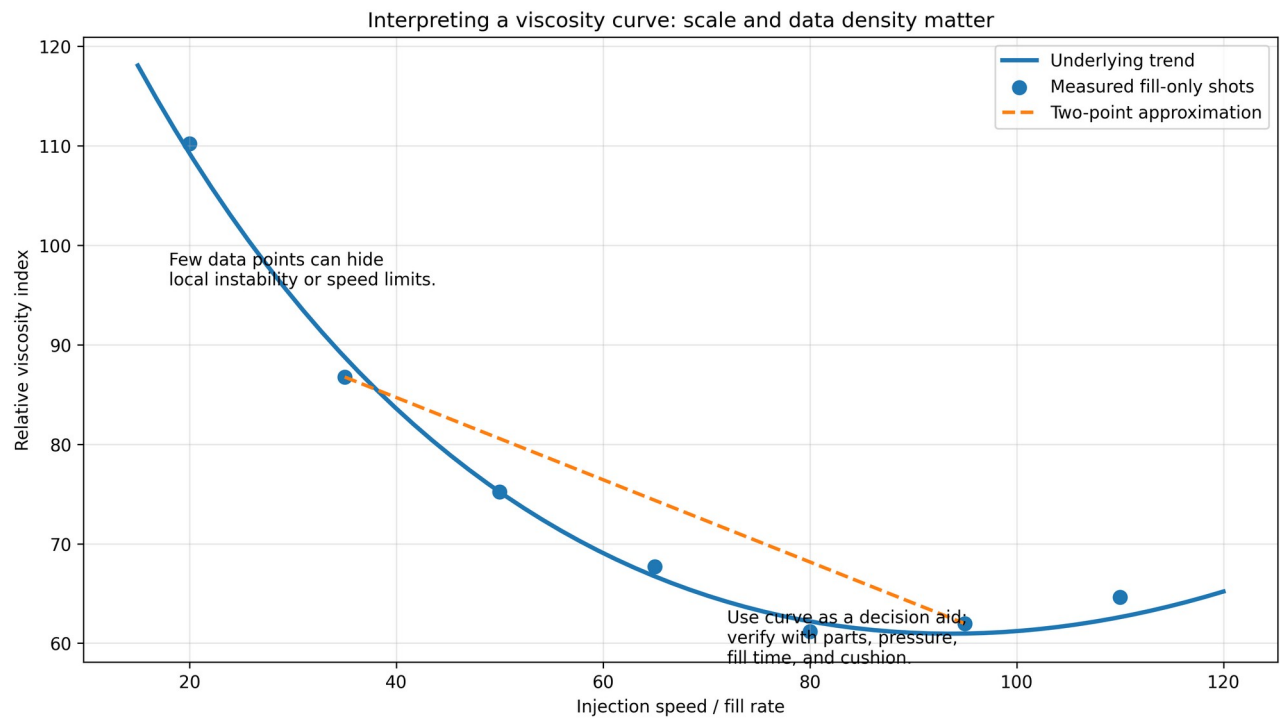


Figure 3. Viscosity-curve interpretation warning: graph scale, data-point count, and response variable selection can change the apparent “flat” region.

1.2 What the study is not

- It is not merely increasing shot size until the part looks full. That practice may create an overpacked process with hidden machine or mold defects.
- It is not the same as a relative-viscosity curve. The viscosity curve evaluates fill-rate sensitivity. Shot size/cushion study evaluates fill volume, V/P transfer, and pressure-transmitting residual melt pad.
- It is not a gate seal study. Gate seal evaluates hold time and part-weight stabilization after the fill and pack settings are already rational.
- It is not a universal cushion rule. The same final-cushion distance can represent a very different melt volume on different screw diameters.

1.3 Required measurements

Measurement	Preferred source	Minimum acceptable source	Notes
Screw position at start of injection	Machine shot-end/recovery data export	Machine screen record	Record with decompression status and screw-diameter context.
Transfer position	Machine data export or setup sheet	Controller screen	Use the controller unit exactly as displayed; do not mix mm, in, cm ³ , and percent without conversion.
Final cushion	Cycle trace, not only final HMI snapshot	Displayed cushion after hold	Record minimum cushion and final stabilized cushion if the controller supports both.
Fill time	Machine process monitor	Stopwatch only for rough check	Fill time is a process response; injection-speed setting is only an input.
Peak pressure	Specific plastic pressure, nozzle pressure, or cavity pressure	Hydraulic pressure with intensification ratio noted	Pressure type must be recorded. Hydraulic pressure and plastic pressure are not interchangeable.
Part weight	Calibrated scale, cavity-separated when possible	Total shot weight	For cold runners, record runner/part separately when diagnosing feed imbalance.
Water and thermal data	Flow meters, inlet/outlet temperature, thermolator and chiller data	Setpoint only	Setpoint alone does not prove heat-transfer capacity.
Hot-runner data	Zone actual temperature, alarms, output %, valve timing,	Controller setpoints	Hot runner setpoints are not adequate proof of tip/manifold thermal

Measurement	Preferred source	Minimum acceptable source	Notes
	sequence data		balance.

2. Root Causes of Why a Shot Size / Cushion Study Is Done

A plant performs this study whenever it must separate fill, pack, and hold behavior, establish a repeatable process, recover a drifting process, transfer a mold to another press, validate a new resin lot, troubleshoot short shots or flash, or prove that the injection unit and mold can support the required process window. The study is especially valuable when the same defect can be caused by more than one subsystem: material viscosity, gate restriction, non-return valve leakage, blocked vents, hot-runner imbalance, water-flow instability, or excessive clamp force.

Trigger	Likely question answered by the study	Cold runner emphasis	Hot runner emphasis
New mold startup	What fill-only shot volume and transfer point create a stable non-packed short shot?	Account for sprue, runner, runner puller, cold slug well, runner imbalance.	Account for manifold thermal state, tip heat, valve timing, gate vestige, startup drool/freeze.
Mold transfer to another press	Does the new screw diameter, check ring, response time, and pressure capability require a new position-to-volume conversion?	Runner and sprue weight may dominate small-part tools, making screw-size matching critical.	Large heated melt inventory may mask or amplify press response differences.
Short shots or drifting cushion	Is the process running out of melt, leaking past the check ring, pressure limited, or losing heat?	Cold runner freeze-off, runner sizing, and gate restriction show up in short-shot progression.	Tip freeze, valve pin timing, heater/T/C failures, and manifold imbalance may mimic material shortage.
Flash after speed increase	Is transfer late, clamp insufficient, vents too deep/damaged, or parting line compromised?	Cold runner pressure loss may spike at sprue/runner/gate transitions.	Valve-gate timing and tip/gate wear can create local flash even if total cushion looks stable.
Medical/automotive validation	Can the process produce statistically repeatable fill, cushion, weight, and dimensions?	Runner regrind policy must be locked down or excluded.	Hot-runner controller, valve-gate sequence, and thermal soak must be part of validation.
Material lot change or regrind change	Did viscosity or melt density change enough to shift transfer, pressure, cushion, or pack response?	Runner regrind can immediately change melt history and feed consistency.	Hot runner residence time makes material transitions slower and more deceptive.

3. Root Causes and Study Drivers

3.1 Material factors

The material system determines how much pressure is required to deliver a given flow rate, how fast the melt freezes in the gate and cavity, how much pack compensation is required, and how sensitive the result will be to temperature, moisture, shear, fillers, and residence time. Polymers are generally non-Newtonian and shear-thinning under injection molding conditions: as shear rate increases, apparent viscosity usually decreases until a practical stable region is reached. The study must therefore be tied to a selected fill rate rather than to a generic speed percentage.

Material factor	Effect on study result	Process implication
Viscosity / MFR / MVR	High-viscosity grades require more pressure and may show a larger cushion/pressure response for the same stroke change.	Select fill speed with a viscosity curve; verify that the press has pressure reserve.
Melt temperature	Higher melt temperature usually lowers viscosity and increases flow length, but too much heat increases degradation and color/odor risk.	Measure actual melt temperature with an approved purge-shot method; do not rely only on barrel setpoints.
Mold temperature	Hotter molds usually delay freeze-off and reduce molded-in stress; colder molds can raise pressure demand and exaggerate short shots.	Use water-flow and inlet/outlet data. Mold-temperature setpoint alone is not enough.
Moisture	Hydrolysis, splay, viscosity loss, brittleness, bubbles, and inconsistent cushion can appear in hygroscopic materials.	Verify dryer temperature, dew point, residence time, and material-handling seals.
Fillers and reinforcements	Glass, mineral, carbon, and impact modifiers change viscosity, shear heat, wear, orientation, shrinkage, and check-ring life.	Expect different pressure-loss and cushion response than the neat resin. Gate and venting may need redesign.
Regrind and contamination	Can shift viscosity and melt density shot-to-shot; may interfere with non-return valve seating.	Lock regrind percentage during the study or run virgin-only study first.
Color concentrate / additives	Can lubricate, thicken, degrade, or plate out; may change hot-runner drool or gate freeze behavior.	Use production-approved letdown ratio and supplier-approved carrier compatibility.

3.2 Polymer-specific expectations: PP, ABS, PC, and related materials

Polymer family	Expected influence on shot size/cushion study	Cold runner notes	Hot runner notes
PP (polypropylene)	Semi-crystalline; usually easier flow than PC, higher shrink/pack demand, sensitive to cooling and crystallization. Cushion trends may look stable while dimensions move with mold temperature.	Runner regrind and cold-runner weight can be a large part of total shot. Gate seal may occur quickly in small gates.	Hot runners are common; thermal balance affects crystallization, gate vestige, stringing, drool, and cavity balance.
ABS	Amorphous; cosmetic-sensitive; viscosity and surface quality respond strongly to melt/mold temperature and fill rate. Drying discipline still matters.	Cold slug wells and runner design affect splay, blush, knit-line appearance, and pressure loss.	Hot runner shear and residence can affect color, streaks, and gate blush; use conservative thermal profiles.
PC (polycarbonate)	High viscosity, high processing temperature, moisture-sensitive; narrow degradation window. Cushion excess can increase residence damage.	Sprue/runner/gate pressure loss can be severe; oversized runners may be needed for high optical or structural parts.	Requires uniform heat and low dead spots; tip freeze or overheated tips can shift shot/cushion conclusions.
PC/ABS	Intermediate behavior; often used in automotive/consumer housings. Melt temperature and fill speed affect residual stress and weld-line strength.	Runner and gate sizing often dominate pressure drop in large cosmetic parts.	Hot-runner thermal balance and sequential valve timing can change apparent fill pattern.
PA / nylon	Hygroscopic; moisture and drying status can change viscosity, flash, brittleness, and cushion repeatability.	Runner regrind moisture pickup can be significant during long exposure.	Residence and moisture-related degradation can be amplified by manifold volume.
TPE/TPV/soft materials	Compressible and often shear/temperature sensitive; small cushion may be recommended by material supplier, but validate on tool.	Cold runner pull and recovery can cause cycle-to-cycle variation.	Drool/stringing/tip shutoff and valve timing are central to cushion interpretation.

3.3 Process parameters

Parameter	If increased	If decreased	False conclusion risk
Injection speed / fill rate	Usually lowers apparent viscosity by shear thinning until mechanical, venting, or degradation limits are reached. Can create burns, jetting, blush, flash, or high clamp demand.	Can increase viscosity, cause premature freeze, flow lines, short shots, and poor weld-line strength.	Choosing cushion after a slow fill may require a larger shot than the production fill speed needs.
Injection pressure limit	Allows velocity control to be maintained if high enough. It does not force the mold to fill unless speed and transfer are set correctly.	Can pressure-limit the shot, producing false short shots and misleading cushion.	A pressure-limited fill-only study is invalid for transfer and viscosity conclusions.
Transfer position	Later transfer packs during fill and reduces cushion; can flash or hide gate/vent issues.	Earlier transfer leaves more cushion but may underfill before pack.	Part may look full due to pack, while fill-only state is wrong.
Hold pressure	Increases packing, part weight, dimensions, stress, flash risk, and cushion consumption.	May create sink, voids, shrink, short dimensions, and low part weight.	Cushion loss during hold may be normal pack or check-ring leakage; compare part weight.
Hold time	Increases weight until gate seal, then adds no useful packing and may waste cycle time.	May release pressure before gate seal, causing weight/dimension variation.	Calling the test “gate freeze” can mislead; it is a hold-time/part-weight stabilization study.
Melt temperature	Reduces viscosity but may increase degradation, drool, flash, and residence sensitivity.	Raises viscosity, pressure, and short-shot risk; may improve some dimensional targets.	Barrel setpoint is not actual melt temperature.
Cooling rate / mold temperature	Higher mold temperature can improve flow and surface but extend cooling and change shrinkage.	Lower mold temperature can freeze gates sooner and require more pressure/shot.	Water-flow instability may be mistaken for material or machine variation.

3.4 Gate design, size, and location

Gate geometry is often where a shot size/cushion study exposes the difference between “the machine moved plastic” and “the cavity received plastic.” A small or poorly located gate can generate high shear heat, high pressure drop, unbalanced cavity filling, gate blush, freeze-off before adequate packing, and unstable cushion consumption. A large gate can extend pack control and reduce pressure loss but can

cause vestige, long gate seal time, stringing, drool, and trimming problems. Gate location affects flow length, weld-line position, air-trap locations, fiber orientation, stress, and fill balance.

4. Mold Design: Cold Runner vs Hot Runner Effects

Cold runner vs hot runner: what the shot-size/cushion study actually see

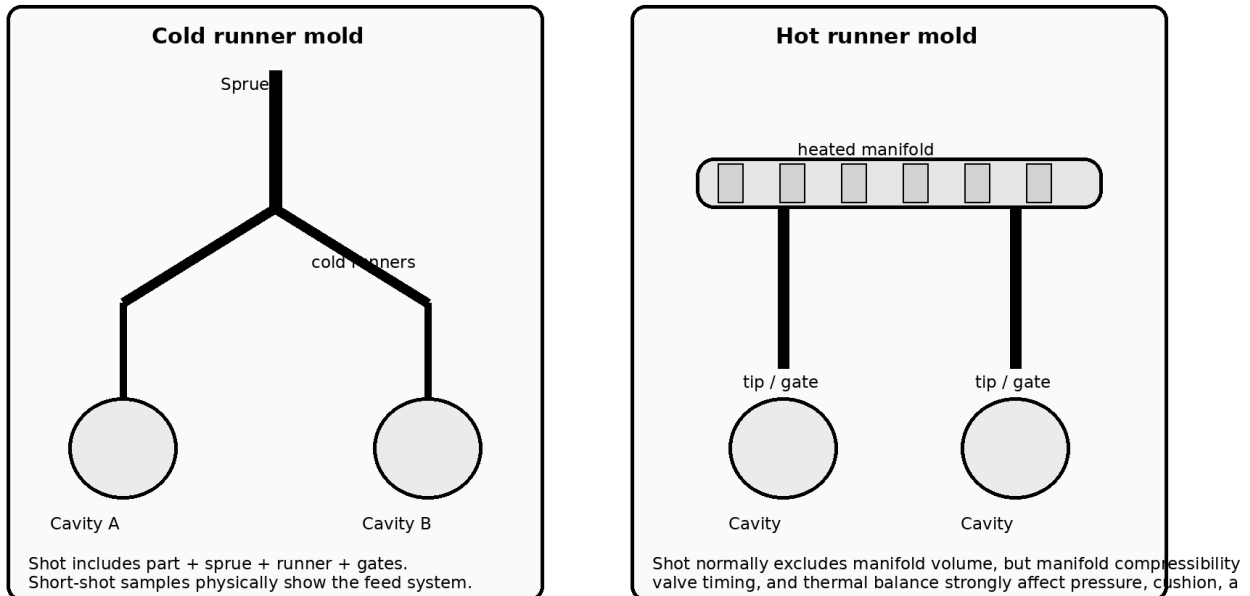


Figure 4. Cold runner and hot runner systems produce different shot-volume, pressure-loss, residence, and diagnostic behavior.

4.1 Cold runner molds with sprues and runners

In a cold runner mold, every cycle injects and ejects the part plus the sprue, runners, sub-runners, cold slug wells, and gates. The cold runner system can be a large percentage of total shot size, especially on small parts. Therefore, cushion and shot-size decisions are sensitive to runner weight, runner temperature, cold-slug behavior, sprue-bushing contact, nozzle-seat sealing, runner balance, and regrind practice.

- Short-shot studies are visually informative because the sprue, runner, gate, and cavity progression are physically ejected and can be inspected.
- Pressure-loss studies can progressively isolate nozzle, sprue, runner, gate, and cavity restrictions, but machine-only pressure readings can mislead if not interpreted carefully.
- Runner regrind changes the material stream; the study should document whether the trial was virgin-only, regrind-included, or production mix.
- Cold runner tools are sensitive to sprue sticking, cold slugs, runner pull, runner imbalance, gate freeze, and water-line imbalance around the runner plate.

4.2 Hot runner molds with tips, manifolds, and valve gates

In a hot runner mold, the manifold and drops are heated and normally remain full between shots. The ejected shot is usually the part plus a gate vestige or small cold sub-runner, not the entire manifold volume. However, the manifold volume remains a compressible, heated melt reservoir between the screw and cavities. That reservoir can damp pressure signals, delay response, store degraded material, hide imbalance, and change apparent cushion behavior.

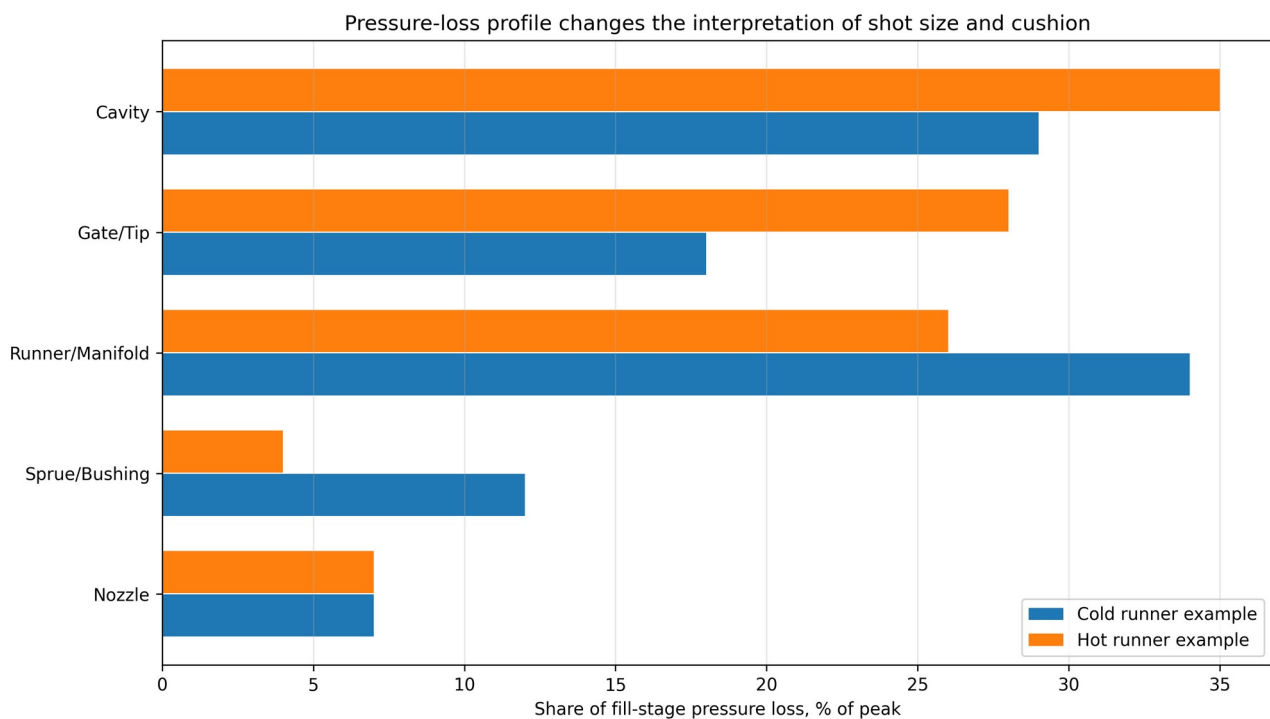
- Tip and manifold temperatures must be verified as actual values and alarm history, not just setpoints. A single weak thermocouple, heater band, or output-limited zone can distort study conclusions.
- Valve-gated systems add valve timing, stem movement, pneumatic/hydraulic/electric actuation health, stem wear, and gate shutoff quality to the shot/cushion problem.
- Hot-runner imbalance can be thermal, rheological, geometric, or timing-related. A balanced CAD manifold can still produce unbalanced short shots if tips, gates, valve pins, or local cooling are not balanced.
- Material changes and color changes have longer stabilization periods because old material remains in the hot-runner melt channels. Do not run a cushion study during transition unless that transition is the object of the study.
- Hot runner molds require extra safety discipline. Never bypass guarding, open electrical cabinets, or loosen hot-runner components under heat/pressure. Follow OEM lockout/tagout and depressurization procedures.

Factor	Cold runner interpretation	Hot runner interpretation
Shot volume	Includes cold feed system every cycle.	Normally excludes heated manifold volume, but manifold compressibility affects response.
Cushion drift	Can indicate check-ring leakage, nozzle leak, packing, cold runner freeze change, or feed inconsistency.	Can also indicate tip freeze, valve timing variation, manifold thermal drift, or hot-runner leakage/drool.
Short-shot samples	Runner and cavity fill progression are directly visible.	Cavity fill is visible; manifold internal distribution is not visible without sensors/simulation.
Pressure loss	Sprue and runner restrictions are physically inspectable and measurable by progressive short shots.	Manifold, drop, tip, and gate pressure loss may require simulation, sensor data, or supplier analysis.
Residence time	Runner is ejected; barrel residence dominates for material in the machine.	Barrel plus manifold/drop residence must be considered. Dead spots can invalidate cushion conclusions.
Balance tuning	Steel changes, runner/gate changes, water balance, vent corrections.	Zone temperature changes, valve timing, tip/gate service, manifold balance, plus steel changes when needed.
Failure modes	Sprue sticking, runner hang-up, cold slug, gate freeze, runner	Tip freeze/drool, valve pin sticking, manifold leak, heater/T-

Factor	Cold runner interpretation	Hot runner interpretation
	imbalance.	C fault, sequencing error, material degradation in drops.

4.3 Wall thickness variation and venting

Wall thickness determines local flow resistance, cooling rate, pressure transmission, and pack response. Thin sections can freeze before the flow front reaches the end of fill, forcing higher shot size or pressure and causing false “not enough material” diagnosis. Thick sections continue to accept pack longer and can consume cushion during hold. Abrupt thick-to-thin transitions create hesitation, weld lines, trapped air, and internal stress. Proper venting is mandatory because air has to leave at the same rate plastic arrives; otherwise burns, dieseling, short shots, and transfer-position shifts may be misdiagnosed as viscosity problems.



Illustrative. Use short-shot pressure-loss study, nozzle/cavity sensors, or simulation to quantify actual tool-specific losses.

Figure 5. Example pressure-loss profile. The “right” shot and cushion depend on where the system loses pressure.

5. Machine Factors and Auxiliary Equipment

5.1 Screw design, screw/barrel wear, non-return valve, and back pressure

The injection unit is the measuring instrument and actuator for a shot size/cushion study. If the screw, barrel, check ring, check-ring seat, ball check, mixing section, feed throat, hydraulic/electric drive, or position transducer is unstable, the study may produce a polished but false result. The most common machine-side failure is non-return-valve leakage or variable seating, which changes the amount of melt delivered to the mold for the same commanded screw movement.

Machine condition	Observable during study	Likely effect	Recommended verification
Worn check ring / NRV	Cushion drifts downward under hold, shot-to-shot part weight changes, peak pressure varies.	Material leaks back over screw during injection/hold.	Static and dynamic check-ring tests; trend cushion, fill time, peak pressure, part weight.
Worn screw/barrel	Poor recovery repeatability, unstable melt quality, black specks, long recovery, cushion variation.	Reduced pumping efficiency and inconsistent melt density.	Measure screw/barrel clearances during PM; compare recovery time and torque.
Incorrect screw design	Poor melting, unmelt, excessive shear heat, long recovery, degradation.	Melt homogeneity shifts shot-to-shot.	Review L/D, compression ratio, mixing section, resin family, filler abrasiveness.
Back pressure too low	Inconsistent recovery, air entrapment, poor mix, unstable shot volume.	Shot mass and melt density vary.	Increase only enough to stabilize melt quality without excessive heat/recovery time.
Back pressure too high	Long recovery, high melt temperature, degradation, drool, cushion instability.	Over-shearing and heat history obscure true fill requirement.	Measure actual melt temperature and recovery time.
Position transducer / calibration issue	Transfer and cushion values repeat poorly or disagree with actual movement.	Data are not trustworthy.	Calibrate/verify screw-position system and zero/bottom position.

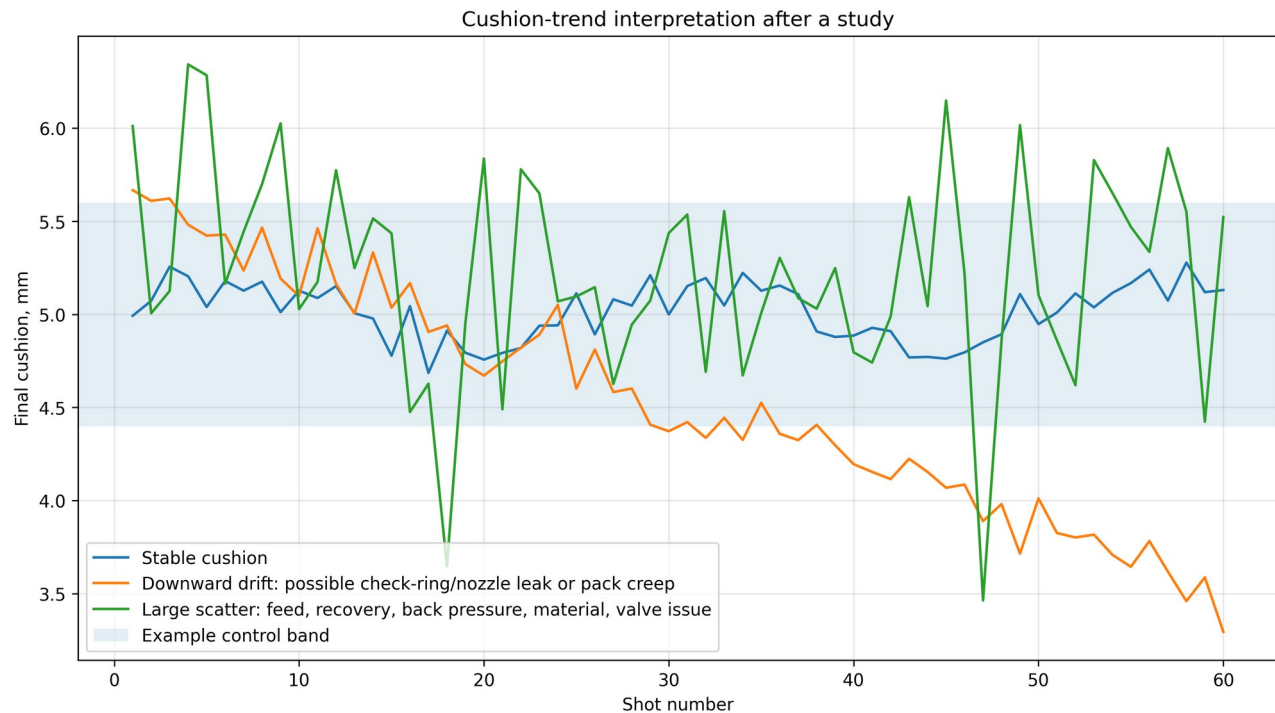


Figure 6. Cushion trend patterns after the study. Stable mean and low scatter are more important than a single screenshot value.

5.2 Clamping force and mold protection

Clamp force does not fill the part. Clamp force keeps the mold closed against cavity pressure. In a shot size/cushion study, inadequate clamp force can cause flash, apparent full parts at insufficient shot, parting-line damage, and false interpretation of viscosity or shot volume. Excessive clamp force can crush vents, trap gas, increase burns, accelerate mold wear, and hide venting problems. Clamp must be sufficient, documented, and tied to projected area and actual cavity pressure, not simply set to maximum.

5.3 Auxiliary equipment: chillers, thermolators, hot-runner controllers, robots, sprue pickers

Auxiliary	Role in the study	Failure mode that creates false conclusions
Chiller / thermolator / TCU	Controls mold temperature and freeze pattern. Determines gate seal, pressure demand, shrink, and ejection stability.	Low flow, clogged circuits, wrong plumbing, unstable inlet temperature, bad pump, poor heat transfer.
Hot-runner temperature controller	Controls manifold/drop/tip heat, drool, freeze, fill balance, material degradation, and valve-gate quality.	Bad thermocouple, wrong zone map, heater failure, output saturation, poor grounding, wrong standby/startup sequence.
Material dryer and loader	Controls moisture, feed consistency, material residence, contamination, and regrind ratio.	Wet resin, bridging, feed starvation, fines, wrong material, unstable regrind ratio.
Robot / sprue picker	Removes parts/runners consistently and protects short-	Part hanging, mold-close delay variation, runner not clearing,

Auxiliary	Role in the study	Failure mode that creates false conclusions
	shot samples during study.	short shots damaged before inspection.
Grinder / regrind system	If used in production, it changes feed bulk density, moisture, and heat history.	Variable granule size, fines, contamination, uncontrolled ratio, warm material returned too quickly.
Conveying and central material system	Maintains correct resin and steady feed to throat.	Wrong line purged into machine, poor dedusting, leaking vacuum line, material cross-contamination.

6. Real-World Examples and Case Studies

The following case studies are anonymized, field-realistic technical scenarios synthesized from common scientific-molding practice and public practitioner discussions. They are not presented as proprietary customer data. Use them as decision templates, not as proof that the same root cause applies in your plant.

6.1 Case Study A - PP medical closure, cold runner, 24 cavities

Problem: A polypropylene child-resistant closure tool showed intermittent short shots on the last-filling cavities after a resin lot change. Operators increased shot size and hold pressure, which restored visual fill but produced high removal torque variation and occasional flash at the parting line.

Study method: The team stabilized resin drying and water circuits, disabled hold pressure for fill-only shots, used the established viscosity curve to select the fill speed, and reduced shot/transfer until the best-filling cavity was approximately 95-98% full. Runner and cavity short shots were saved by shot number. Final cushion, fill time, peak pressure, total shot weight, and cavity-separated part weights were recorded for 30 shots.

Findings: The cold runner was partially balanced, but two sub-runners had higher pressure loss and corresponding low fill. Cushion was stable, which reduced suspicion of check-ring leakage. A pressure-loss review showed excessive runner/gate contribution. Vents at the end of the weak cavities were contaminated.

Resolution: Clean vents, polish runner restrictions, re-establish transfer, then run gate seal study. The final process used slightly higher fill speed in the stable viscosity region, lower hold pressure, and a narrower cushion alarm band. Scrap was reduced without carrying excessive shot size.

6.2 Case Study B - PC/ABS automotive bezel, hot runner valve gate

Problem: A cosmetic PC/ABS bezel had alternating gate blush and short shot near a remote rib. The press cushion looked acceptable, so the initial process reaction was to increase melt temperature and shot size. The change reduced short shots but increased splay and odor.

Study method: The team treated the hot runner as part of the measurement system. They verified zone mapping, actual tip temperatures, valve-gate timing, and manifold soak. Fill-only studies were run with

valve timing locked and then with a sequenced profile. Cavity pressure sensors were not available, so nozzle pressure, screw position, fill time, part weight, and short-shot photographs were used.

Findings: One valve gate opened late because of pneumatic actuation restriction. The late gate caused a false shot-size conclusion: more material was being commanded, but the local cavity was starved due to timing. The final cushion was acceptable because the manifold and other cavities accepted pack.

Resolution: Restore valve actuation, clean the tip, correct timing, and lower melt temperature back into the supplier range. Re-run fill-only transfer study, then pack/hold study. The final process used less shot than the temporary “fix,” lower degradation risk, and improved appearance.

6.3 Case Study C - ABS consumer housing, cold runner, poor venting

Problem: A thin-wall ABS housing flashed at one corner while also showing burn marks and incomplete fill at a far rib. Operators alternated between more shot and less shot, producing unstable visual quality.

Study method: A relative-viscosity curve was run first and showed a stable fill-rate region below the fastest available speed. During shot/cushion study at that fill rate, the tool made burn marks before reaching the intended fill-only percentage.

Findings: The problem was not initial shot shortage. It was air evacuation and local parting-line support. Excessive clamp force had reduced vent effectiveness, while local mold support allowed flash under pressure.

Resolution: Clean and correct vents, verify clamp, repair local shutoff/support, then re-run the study. After venting was corrected, required pressure dropped, cushion stabilized, and the transfer point could be set without overpacking.

6.4 Public practitioner accounts and how to interpret them

Public forums contain useful firsthand observations, but they are uncontrolled evidence. Treat them as practitioner signals, not as validated procedures. One Reddit processor described keeping part weight nearly constant across speed changes and using percent change in viscosity to choose a run point. Another warned that changing shot size incorrectly during a viscosity study can make the curve invalid. An Eng-Tips discussion tied cushion loss during cooling/hold to either legitimate packing or a leak, recommending part-weight comparison to separate those cases. These comments align with good practice: preserve the study objective, control fill-only conditions, and compare cushion movement with part weight and pressure rather than interpreting cushion alone.

7. Diagnostic Techniques

7.1 How to identify whether a study has already been done

A molded part alone cannot prove that a shot size/cushion study was performed. Visual inspection, microscopy, and metrology can show outcomes consistent with correct or incorrect fill/pack settings, but they cannot prove the process-development method. To determine whether the study was done, review the process-development package.

Evidence to look for	Strong evidence	Weak evidence
Setup sheet	Recorded transfer position, fill time, peak pressure, cushion band, fill-only weight, final part weight.	Only final shot size and hold pressure.
Trial samples	Labeled short shots at selected fill rate showing 95-98% fill target and cavity balance.	Random incomplete parts with no shot number or settings.
Data log	At least 10-30 consecutive shots with screw position, cushion, pressure, fill time, and weight.	Single screenshot of cushion.
Study sequence	Viscosity/fill-rate study, transfer study, pressure-loss/cavity-balance checks, gate-seal/hold-time study.	Operator note: “added material until full.”
Hot runner record	Zone actuals, alarms, output %, tip/valve service status, valve timing, soak time.	Only nominal hot-runner setpoint list.
Quality confirmation	Dimensional and weight capability run tied to final process window.	Only first-piece approval.

7.2 Visual, microscopic, and destructive inspection

Method	Can reveal	Cannot prove
Short-shot visual progression	Flow hesitation, last-fill locations, imbalance, weld-line formation, air traps, gate freeze clues.	Exact cushion adequacy without machine data.
Gate/runner inspection	Gate blush, jetting, shear burn, cold slug, runner hang-up, sprue sticking.	Hot-runner manifold condition unless disassembled/inspected.
Microscopy / microtome section	Voids, weld-line bonding, fiber orientation, unmelt, contamination, internal stress features.	That transfer position was scientifically established.
Polarized-light inspection on transparent parts	Residual stress/birefringence patterns.	Root cause without process data.
Part weight by cavity	Pack consistency, balance, gate seal stability.	Whether short shot is caused by machine, material, mold, or HR without additional data.
Dimensional capability	Overpack/underpack, cooling imbalance, shrink sensitivity.	Correct shot size unless tied to fill-only and cushion records.

7.3 Mold-flow analysis and simulation

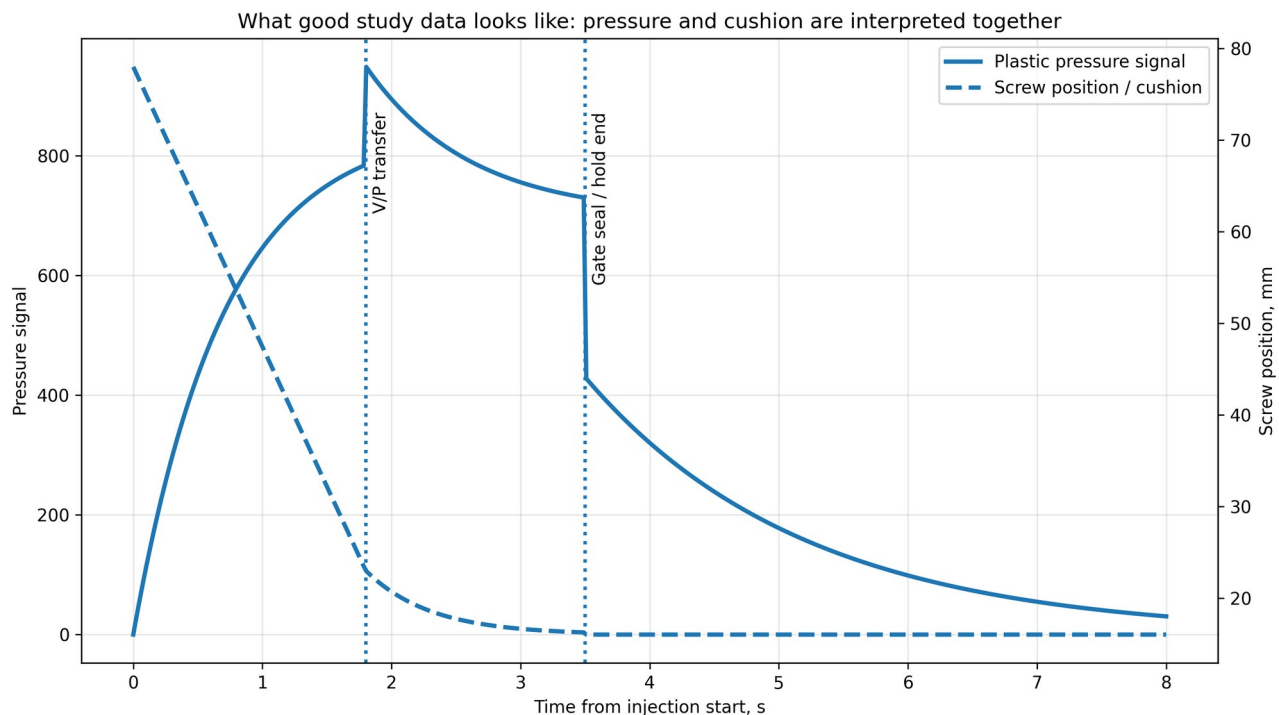
Mold-flow software is useful before steel changes and before machine trials, but it should not replace a shop-floor shot size/cushion study. Simulation can predict fill pattern, pressure drop, air traps, weld lines, shear rate, shear heating, clamp force, gate freeze tendency, cooling imbalance, and hot-runner thermal

issues. It cannot know the actual condition of a worn check ring, blocked vent, miswired hot-runner zone, contaminated resin, or damaged gate unless those conditions are modeled or measured.

Simulation output	How it helps the study	Validation needed at press
Fill time / flow front	Predicts last-fill areas and likely short-shot progression.	Compare with actual short-shot samples.
Pressure drop	Identifies whether nozzle, sprue, runner, gate, manifold, tip, or cavity dominates.	Compare with pressure-loss study or sensor data.
Shear rate / shear heating	Flags gates, tips, thin walls, or hot runner zones that may degrade material.	Inspect gate blush, burn, melt temperature, color, mechanical performance.
Air traps / venting	Guides venting and vacuum strategy.	Inspect burns, gas marks, parting-line gas, vent residue.
Hot runner thermal balance	Identifies temperature variation, dead spots, residence concerns.	Verify HR zone mapping, T/C health, actual tip performance, valve timing.
Packing and shrinkage	Helps choose gate size, hold pressure/time, cooling design.	Confirm with gate seal, part weight, dimensions, cavity pressure if available.

7.4 Alternative and advanced methods

- Cavity-pressure based transfer: Use a cavity-pressure sensor near the gate or end of fill to transfer based on actual cavity condition rather than only screw position. This is stronger but requires sensor-equipped tooling and disciplined calibration.
- Nozzle-pressure based studies: Nozzle pressure gives a closer melt-pressure signal than hydraulic pressure and is useful when cavity sensors are unavailable.
- Tie-bar strain and clamp-force monitoring: Can detect mold separation, clamp-force response, and pressure-related variation, especially in validation or adaptive-control systems.
- Dynamic check-ring testing: Tests whether the non-return valve leaks under realistic injection conditions before blaming the mold or resin.
- Short-shot balance study: Particularly valuable for multi-cavity tools. It shows which cavities fill first, which are pressure-starved, and whether hot-runner or cold-runner balance is acceptable.
- Weight-based adaptive control: Uses part/shot weight or pressure-derived estimates to compensate for viscosity and density shifts. It must be validated against quality characteristics, not only mass.
- Non-destructive testing: Industrial CT can detect voids, sinks, internal weld defects, insert displacement, and some internal flow defects. Ultrasonic methods can detect voids or delamination in suitable geometries. Infrared thermography can expose cooling imbalance. These methods assess consequences; they do not by themselves set shot size.



Illustrative trace. Actual signals may be hydraulic pressure, specific pressure, nozzle pressure, or cavity pressure depending on available instrumentation.

Figure 7. Pressure and cushion traces should be interpreted together; neither signal alone fully describes the molded part.

8. Preventive Measures Before and After Running the DOE

8.1 Material selection and preparation

- Choose a resin grade with a melt-flow range appropriate for flow length, wall thickness, gate type, cosmetic requirement, mechanical requirement, and hot-runner residence. Do not select resin by MFR alone; review supplier processing guidance, drying requirements, shear sensitivity, degradation limits, and additive compatibility.
- For PP closures and packaging, control MFR/MVR, nucleation, slip/antistatic packages, color concentrate, and regrind. For ABS, control drying, cosmetic surface, heat history, and gate blush. For PC/PC blends, control moisture, actual melt temperature, residence time, and shear. For filled materials, account for wear, orientation, and pressure drop.
- Lock material condition before the study: resin lot, dryer conditions, dew point, hopper residence, regrind percentage, color ratio, and material line. A shot/cushion study on unstable material is not a study; it is noise collection.

8.2 Process optimization: what the study tells you

The study tells the processor how much melt the process needs, where fill should end and pack should begin, whether the press can maintain velocity control, whether cushion is adequate and repeatable, and whether pack/hold is consuming cushion in a controlled way. It also tells the processor when a “full part” was achieved for the wrong reason: excessive fill-stage packing, late transfer, high hold pressure, vent restriction, or hot-runner imbalance.

Study result	Interpretation	Optimization action
Fill-only part is below target and cushion is large	Transfer may be too early, shot stroke too small for target, or pressure-limited fill.	Increase shot/late transfer only after confirming pressure limit and fill speed.
Fill-only part exceeds target or flashes before transfer	Transfer too late, shot too large, clamp/vent/shutoff issue, or speed too aggressive.	Reduce shot/earlier transfer, inspect clamp/vents/shutoffs, check speed region.
Cushion reaches zero	The press ran out of melt pad; hold pressure cannot transmit properly.	Increase shot size or transfer earlier only after confirming no overpack/flash; inspect check ring if drift occurs.
Cushion large and variable	Excess melt inventory, poor recovery, back-pressure issue, check-ring instability, material feed variability.	Reduce shot if part data allow; stabilize recovery/feed; inspect NRV.
Peak pressure rises with constant fill time and cushion	Material viscosity, melt/mold temperature, venting, or restriction changed.	Check material lot/dryer/temperature/vents/hot runner before changing shot size.
Hot runner shows cavity imbalance with stable machine signals	Manifold/tip/valve/timing imbalance may be local and hidden from machine cushion.	Perform cavity-separated short-shot/weight study; verify HR zones and valve timing.

8.3 Cold runner vs hot runner process adjustments

Adjustment	Cold runner effect	Hot runner effect
Injection speed	Changes shear in sprue, runner, gate, and cavity; visible in short-shot feed pattern.	Changes shear in manifold, drops, tips, gates, and cavity; may interact with valve timing and tip heat.
Barrel/melt temperature	Changes runner freeze, pressure loss, sprue pull, gate seal, and regrind behavior.	Changes barrel plus manifold residence; may require HR zone adjustments or thermal soak.
Mold temperature	Changes runner/cavity freeze and ejection; can change runner pull and gate seal.	Changes gate/tip cooling and cavity freeze; may shift gate vestige, stringing, or tip freeze.
Transfer position	Directly changes how much feed system and cavity are filled before pack.	Directly changes cavity fill but manifold compressibility may make pressure response less immediate.
Hold pressure/time	Packs part through gate until gate seal; runner/gate freeze governs transmission.	Packs through hot tip/valve gate; gate shutoff timing and tip condition govern transmission.
Multi-stage injection	Use to prevent jetting, reduce shear at gate, push through thin regions, or protect vents.	Use with caution: stage changes may interact with valve gates and manifold pressure waves.

8.4 Mold design improvements

- Optimize gate location to reduce flow length, prevent jetting, locate weld lines in low-stress regions, and allow packing of thick sections before gate seal.
- Size gates and runners to reduce excessive pressure loss without creating unacceptable vestige, long gate seal, drool, or stringing.
- Maintain uniform nominal wall thickness where possible; transition with radii and gradual changes when thickness changes are unavoidable.
- Vent aggressively but safely. Vents must let air/gas escape without flashing. Maintain vent depth, width, land length, and vent cleanliness per material and mold design practice.
- For hot runners, design for melt-channel balance, thermal balance, serviceability, valve-pin alignment, leak prevention, and clean temperature-zone mapping.

8.5 Machine calibration and preventive maintenance

- Verify screw and barrel condition, non-return valve function, nozzle seat, injection pressure calibration, screw-position zero, velocity response, and clamp parallelism before using DOE data for validation.
- Maintain barrel heaters, thermocouples, screw tips, check rings, nozzle tips, feed throat cooling, and hydraulic/electric servo systems.
- Document actual machine capability: screw diameter, maximum injection pressure at selected speed, injection rate, shot capacity, recovery capacity, clamp tonnage, tie-bar spacing, platen support, and controller data export resolution.

9. Corrective Actions

9.1 In-process adjustments

Symptom during study	Do first	Then adjust if confirmed
Short shot with stable cushion and low peak pressure	Check transfer/shot target and fill-only setup.	Increase shot or later transfer; verify not pressure-limited.
Short shot with high peak pressure	Check injection pressure limit, gate/runner restriction, venting, melt/mold temp.	Increase fill speed/temp only within material and tool limits; redesign restriction if needed.
Zero cushion after hold	Confirm no data artifact. Check part weight and hold pressure.	Increase shot or reduce pack only if quality allows; inspect check ring if cushion falls irregularly.
Cushion drifts lower over time	Compare weight increase to cushion loss; inspect for nozzle leak and check-ring leak.	Repair NRV/nozzle or correct gate seal/hold time.
Flash during fill-only	Check transfer/shot, clamp, parting line, vent depth, local support.	Reduce fill speed or shot only after confirming mold is mechanically sound.
Burns before full fill	Check vents, trapped air, fill speed, clamp force, gas generation, material degradation.	Improve venting or reduce speed/shear; avoid masking by increasing shot.

Symptom during study	Do first	Then adjust if confirmed
Hot runner cavity imbalance	Check HR zone mapping, actual temps, valve timing, gate/tip condition, manifold balance.	Tune zones/timing only within supplier guidance; service tips/valves or redesign if needed.

9.2 Use of hot runner systems to improve melt flow

A hot runner can reduce cold-runner waste, reduce required shot weight, improve cycle efficiency, and reduce pressure loss when designed correctly. It can also create new failure modes: dead spots, thermal imbalance, tip freeze, drool, valve pin wear, manifold leakage, and extended residence time. Hot runner adjustments are corrective actions only when the hot runner is the verified constraint. Do not use hot-runner temperature manipulation as a substitute for fixing gate size, venting, valve timing, or material drying.

9.3 Post-processing solutions: cold runner and hot runner molds

Post-processing does not correct an incorrect shot size/cushion study; it only manages downstream effects. Polishing may remove superficial flow marks but will not restore internal weld-line strength. Painting can hide cosmetic variation but may fail over molded-in stress or contamination. Machining or trimming can correct a gate vestige but not overpack stress. Annealing can reduce stress in selected materials but adds cost and may change dimensions. Use post-processing only when validated as part of the product specification, not as a routine escape from process instability.

9.4 Redesign strategies

- Modify part geometry to remove avoidable thin choke points, abrupt wall transitions, isolated thick sections, and flow-front traps.
- Relocate or add gates to reduce flow length and balance pressure delivery. In hot runner tools, consider sequential valve gating when weld lines, pressure, or cosmetics require controlled flow-front management.
- Increase runner/gate cross section where pressure-loss study or simulation shows excessive restriction; reduce where shear, vestige, or gate seal is unacceptable.
- Improve venting at actual last-fill locations identified by short shots and simulation. Do not vent based only on CAD intuition.
- Collaborate with mold makers and hot-runner suppliers using actual pressure, short-shot, temperature, and part-weight data instead of subjective “it needs more shot” claims.

10. Industry Best Practices

Best practice is to define the process by plastic behavior, not only machine settings. This means recording actual fill time, actual pressure response, fill-only state, transfer, cushion, part weight, dimensions, water data, and tool-specific thermal state. Standards such as ASTM D3641 and ISO 294 emphasize reproducible molding conditions for test specimens and the importance of defining the main molding parameters; they do not prescribe a universal production cushion. Machine and hot-runner OEMs increasingly support adaptive controls, pressure/screw-position monitoring, and intelligent assistance systems, but those systems still require a physically sound baseline process.

Best-practice rule	Reason
Run the viscosity/fill-speed study before final transfer and cushion whenever the process is new or materially changed.	A transfer target is only meaningful at a defined fill rate.
Use fill-only shots to separate fill from pack.	Hold pressure can hide an incorrect shot size.
Record pressure type and units.	Hydraulic, specific plastic, nozzle, and cavity pressures are different signals.
Use a cushion control band rather than a single target.	Shot-to-shot process control depends on variation, not one value.
Never accept zero cushion in conventional molding unless a validated special process explicitly requires a different method.	With no melt pad, pressure transmission and machine protection are compromised.
For cold runners, separate runner and part weights when diagnosing shot size.	Runner variation can hide cavity variation.
For hot runners, include zone actuals, valve timing, and soak state in the study record.	Hot-runner state can dominate fill balance and cushion response.
Preserve study samples and records.	Future technicians need evidence, not folklore.

11. Emerging Technologies and Research

11.1 Smart molding and IoT-enabled monitoring

Modern presses and auxiliary systems can capture screw position, injection pressure, injection speed, back pressure, torque, cushion, recovery time, barrel temperatures, water temperatures, flow, hot-runner zone actuals, valve timing, clamp force, tie-bar strain, and alarms. The useful implementation is not data accumulation; it is a closed feedback loop with alarm limits tied to quality-critical outputs. For a shot size/cushion study, smart molding should preserve the baseline curve, monitor deviations, and prevent unapproved operator compensation that hides root cause.

11.2 AI-driven process optimization

AI and adaptive control can assist with transfer point, hold pressure, shot weight, and viscosity compensation by comparing current cycle signals with a reference cycle. These systems are useful for normal variation such as lot-to-lot viscosity shifts, but they cannot legally or technically justify ignoring mold damage, vent blockage, wrong material, failed dryer, failed hot-runner zone, or worn check ring. AI should be used as supervised process control with engineering limits, audit trails, and quality validation.

11.3 Material science innovations

New resin systems, nano-additives, recycled-content streams, self-healing materials, conductive fillers, bio-based polymers, and highly filled compounds can alter viscosity, thermal conductivity, shrinkage, pressure demand, screw wear, and hot-runner residence behavior. These materials increase the value of a disciplined shot size/cushion study because legacy settings and generic cushion rules become less reliable as material behavior diverges from conventional commodity resins.

12. Troubleshooting Guide and Shop-Floor SOP

Shot size / cushion study workflow - cold and hot runner checkpoints

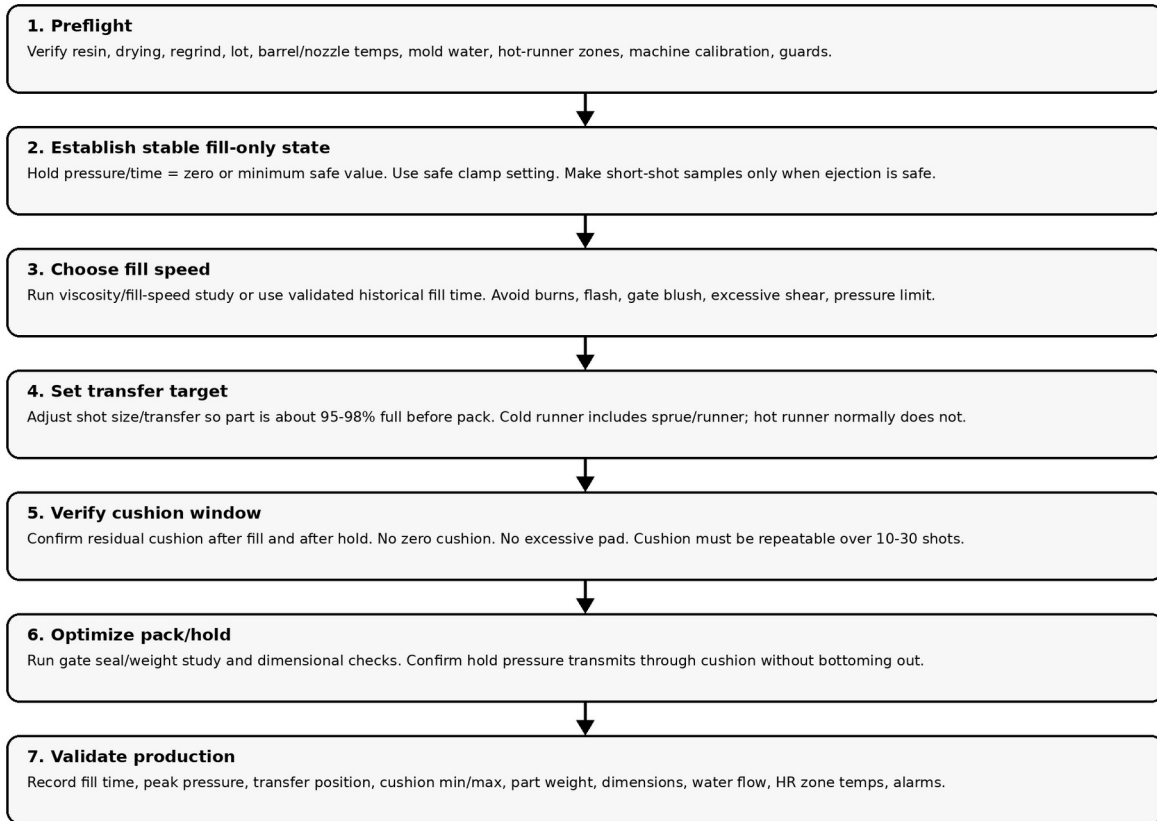


Figure 8. Step-by-step workflow for running and validating a shot size/cushion study.

12.1 Prerequisites before running the study

Prerequisite	Minimum requirement	Why it matters
Safety	Guards functional; lockout/tagout for maintenance; purge shields; heat gloves/face protection; hot-runner OEM procedure followed.	The study involves hot plastic, high pressure, mold motion, and possible short-shot ejection hazards.
Known material condition	Correct resin, color, regrind, drying, dew point, lot, hopper residence.	Unstable material invalidates the data.
Machine health	No known injection-unit, clamp, heater, position, pressure, or NRV defects.	A sick machine gives false mold/process conclusions.
Mold readiness	Clean vents, clean gates, correct water hookup, no damage, correct mold protection, known ejection reliability.	Short shots can stick or damage mold if not planned.
Hot runner readiness if	Correct zone map, actual	Hot runner state can dominate

Prerequisite	Minimum requirement	Why it matters
applicable	temperatures stable, no alarms, valve timing verified, soak complete.	fill pattern.
Measurement tools	Scale, data sheet, calipers/gages, temperature measurement, water data, sample labels, camera.	Unmeasured studies cannot be repeated or defended.
Environmental state	Stable material room, water supply, ambient changes noted, no major plant utilities instability.	Environmental changes can shift viscosity and cooling during the trial.

12.2 Study sequence

14. **1. Stabilize the cell:** Run until barrel, mold, water, dryer, and hot runner are stable. For hot runners, include soak time and verify no zone is output-limited.
15. **2. Establish safe fill-only conditions:** Set hold pressure/time to zero or minimum safe values per plant practice. Confirm the part can eject as a short shot without damaging pins, slides, lifters, or cores.
16. **3. Select fill speed:** Use the relative-viscosity curve or prior validated fill-time target. Avoid speeds that cause burns, flash, pressure limiting, check-ring instability, or degradation.
17. **4. Adjust transfer/shot to target fill:** At the selected fill rate, adjust shot size and/or V/P transfer until the cavity is about 95-98% full before pack. For multi-cavity tools, make sure the most-filled cavity is not fully packed.
18. **5. Record 10-30 shots:** Record start position, transfer, final cushion, fill time, peak pressure, part/runner weight, short-shot appearance, and alarms.
19. **6. Evaluate cushion:** Confirm the screw never bottoms. Confirm cushion is not excessive. Confirm variation is within plant capability and quality requirements.
20. **7. Optimize pack/hold:** Run hold-pressure and hold-time/gate-seal study. Confirm part weight stabilizes and dimensions meet requirements without consuming all cushion.
21. **8. Validate production process:** Run at least the plant-required qualification lot. Lock alarm limits for fill time, peak pressure, transfer, cushion, recovery, water, and hot-runner zones.

12.3 Common pitfalls and how to avoid them

Pitfall	Why it causes false determination	Avoidance
Running the study before thermal stabilization	Melt viscosity, mold temperature, and hot-runner balance drift during data collection.	Wait for stable actual temperatures, not just setpoints.
Using hold pressure during fill-only setup	Pack masks the true fill volume and transfer point.	Disable or minimize hold until fill-only target is established.
Changing multiple variables at once	Cannot identify causal factor.	Use disciplined one-factor steps or planned DOE.
Pressure-limited fill	Velocity setting is no longer achieved, invalidating viscosity and transfer data.	Set pressure limit high enough for safe velocity control; confirm actual fill time.

Pitfall	Why it causes false determination	Avoidance
Ignoring check-ring behavior	Machine leakage looks like material or mold variation.	Run NRV checks when cushion/weight/pressure scatter is abnormal.
Treating hot-runner setpoint as actual melt condition	Failed heaters/T-Cs or poor zone mapping can dominate fill.	Verify zone actuals, output %, mapping, alarms, valve timing, and soak.
Choosing cushion by generic rule only	Different screw diameters and materials make the same cushion distance non-equivalent.	Use rule as startup boundary, then validate with data and quality.
Saving no samples	Future shifts cannot verify what “95% fill” meant.	Label and retain key short shots and final accepted samples.

13. Forms, Templates, and Acceptance Criteria

13.1 Minimum study data sheet

Field	Entry
Mold / part number	
Press / screw diameter	
Material / lot / color / regrind	
Dryer settings and dew point	
Mold water: inlet/outlet temp and flow	
Hot runner: zone set/actual/output and valve timing	
Fill speed / fill time target	
Injection pressure limit and pressure type	
Shot start position	
Transfer position	
Final cushion target and limits	
Fill-only weight and % fill	
Peak pressure	
Part weight by cavity	
Gate seal hold time	
Final dimensions and quality disposition	
Operator / technician / engineer signoff	

13.2 Suggested acceptance criteria

Acceptance criteria must be defined by product risk, material, mold, customer requirements, and validation level. The following are practical starting points, not universal standards.

Criterion	Typical acceptance logic
No zero cushion	Minimum cushion remains above zero through fill and hold under worst normal material and temperature variation.

Criterion	Typical acceptance logic
Cushion repeatability	Stable over 10-30 consecutive shots with control limits based on material used, machine resolution, and part sensitivity. Some plants use percent-of-material-used bands; others use absolute mm/in bands.
Fill time repeatability	Actual fill time remains within validated process-monitor limits, not merely programmed speed.
Peak pressure reserve	Process does not run at pressure limit during velocity-controlled fill. Maintain appropriate pressure reserve based on machine and validation practice.
Fill-only target	Selected V/P transfer produces approximately 95-98% fill unless the tool/product requires a documented exception.
Cavity balance	Short-shot and packed weights meet internal or customer-defined balance requirements.
Quality outputs	Dimensions, cosmetics, mechanical tests, torque/seal function, and regulatory requirements pass under final settings.

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Appendix A - Interpretation Notes for Training the Next Generation of Molders

A good technician should learn to ask: What did the plastic do? not only What number is on the screen? A shot size/cushion study is successful when a trainee can explain why the selected shot volume, transfer point, and cushion produce a stable molded part and what failure mode would appear if each variable moved high or low.

A good process sheet should contain enough information for another qualified person to reproduce the process on the same cell. It should not rely on tribal memory such as “add a little shot until it looks good.”

The final cushion is a process result, not a magic setpoint. The processor controls screw recovery, decompression, transfer, speed, pressure, hold, temperature, and machine health. Cushion is the remaining evidence after those inputs and the mold/material response have acted.

In cold runner tools, train technicians to inspect the runner and the part. In hot runner tools, train technicians to distrust a stable machine signal until they have also checked cavity balance, valve timing, and hot-runner thermal state.